

Electrical Network Analysis (EE 2004)

Date: February 24th, 2026

Course Instructor(s)

Ms. Sara Kiran

Sessional 1 Exam

Total Time (Hrs):

1

Total Marks:

40

Total Questions:

2

Roll No

Section

Student Signature

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1. Attempt all questions.
2. Attempt all parts of the same question together.
3. Show all the steps with proper circuit diagrams, and answers with proper units. Direct answers are not acceptable.

CLO-1: CLO statement for question Q1 and Q2: Apply circuit analysis techniques to determine the natural and forced response of second order circuits.

Q-1:

[5+5+5+5=20 Marks]

A 0.64 H inductor, a 10 mF capacitor, and a variable resistor are connected in parallel. The resistor is adjusted so that the roots of the characteristic equation are $-10 \pm j7.5$ rad/s. The initial voltage on the capacitor is 4 V and the initial current in the inductor is 0.3 A.

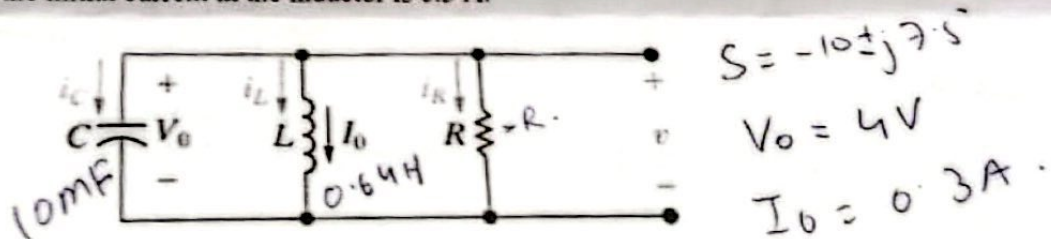


Figure-1

- a. Determine the type of response for this circuit. Justify your answer.
- b. Compute the value of R, and the expression of $v(t)$.
- c. If the value of $R=4\Omega$, and the values of L and C remain the same, determine the type of response for the circuit and compute the roots of the characteristic equations of this response.

Q-2: $\alpha = \frac{1}{2(4)(10m)} =$

[5+5+5+5=20 Marks]

Apply circuit analysis techniques to solve the circuit given in Figure-2. The value of v_g equals $25 \cos(4000t) V$.

- a. Convert the given circuit in frequency domain.
- b. Write down the mesh equations.

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- c. Compute the polar and rectangular expression for V_o and voltage of the dependent source.
d. Determine the steady state expression for $v_o(t) = V_m \cos(\omega t + \phi)$

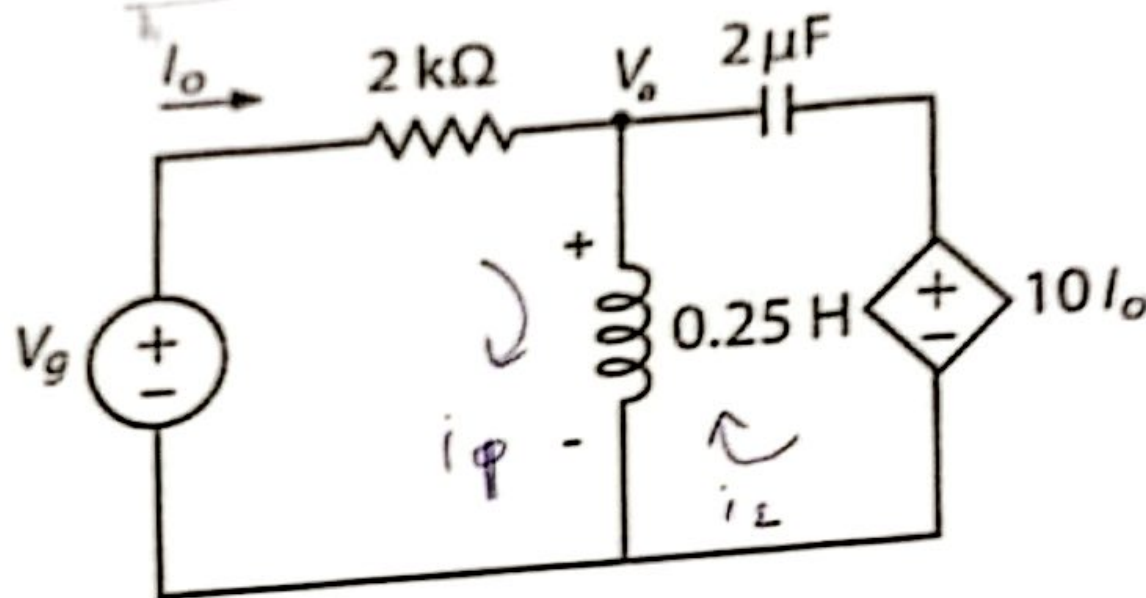


Figure-2

$i_{\phi} = 2.5$